

# PAPER\_561: Buoyancy-Stratified Factorial Geometry — Black Hole Horizon Solution

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**Session:** 149 | **Source:** Composed from CP4 #149, #147 (Sessions 148, 147)

**CP4 Class:** BSFGBlackHoleSolutionHorizonCalculator (#156)

**Date:** 2026-03-27

**Context note:** The BSFG metric  $A_{\mu\nu}(r)$  from CP4 #149 has a time-like component  $A_{00}(r) = 1 + \eta T_{s00}(r) \cos(\pi t_n)$ . A metric horizon occurs where  $g_{tt} = A_{00}(r_h) = 0$ . This paper solves this condition analytically, derives the BSFG surface gravity and Hawking temperature, and contrasts the result with the proplyd equilibrium radius  $r_q$  from PAPER\_550 (CP4 #147).

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## Abstract

This paper presents a UQFF analysis of Stratified Factorial Geometry — Black Hole Horizon Solution, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

### §1 Abstract

We solve the BSFG horizon equation  $A_{00}(r_h) = 0$  and derive:

$$r_h = (\eta C_{\text{num}} |\cos(\pi t_n)|)^{1/3}, \quad (\text{physical when } \cos(\pi t_n) < 0)$$

At  $t_n = 1$  (maximum Aether anti-phase):  $r_h \approx 1.62 \times 10^8 \text{ m} \approx 0.233 R_\odot$ . The BSFG surface gravity and Hawking temperature are:

$$\kappa_{\text{BSFG}} = \frac{3c^2 \eta |C_{\text{num}}| |\cos(\pi t_n)|}{2r_h^4} = \frac{3c^2}{2r_h}, \quad T_H^{\text{BSFG}} = \frac{\hbar \kappa_{\text{BSFG}}}{2\pi k_B c} \approx 3.37 \times 10^{-12} \text{ K}$$

The scale hierarchy:  $r_h \approx 0.23 R_\odot \ll r_q \approx 0.097 \text{ AU}$  — the BSFG horizon (when it exists) lies deep inside the star,  $\sim 145\times$  smaller than the proplyd equilibrium radius.

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### §2 Horizon Condition

The BSFG metric time-time component (CP4 #149):

$$A_{00}(r) = 1 + \varepsilon(r) = 1 + \frac{\eta C_{\text{num}} \cos(\pi t_n)}{r^3}$$

Setting  $A_{00}(r_h) = 0$ :

$$1 + \frac{\eta C_{\text{num}} \cos(\pi t_n)}{r_h^3} = 0 \implies r_h^3 = -\eta C_{\text{num}} \cos(\pi t_n)$$

**Physical requirement:**  $r_h^3 > 0$  demands  $\cos(\pi t_n) < 0$ , i.e.:

$$\frac{1}{2} < t_n < \frac{3}{2} \pmod{2} \quad (\text{Aether anti-phase})$$

In the  $t_n \in [0, 2)$  Aether cycle, a horizon only exists during the anti-phase half. During the normal phase ( $\cos > 0$ ), the Aether is repulsive and no horizon forms.

### §3 Horizon Radius

**Step 1.** At  $t_n = 1$  ( $\cos(\pi t_n) = -1$ ):

$$r_h = (\eta C_{\text{num}})^{1/3}$$

Substituting  $\eta = 10^{-22} \text{ m}^3/\text{J}$  and  $C_{\text{num}} \approx 4.27 \times 10^{46} \text{ J}$ :

$$r_h = (10^{-22} \times 4.27 \times 10^{46})^{1/3} = (4.27 \times 10^{24})^{1/3} \approx 1.62 \times 10^8 \text{ m}$$

**Step 2.** Scale comparisons:

| Length scale                      | Value                           | Ratio to $r_h$              |
|-----------------------------------|---------------------------------|-----------------------------|
| $r_h$ (BSFG horizon)              | $1.62 \times 10^8 \text{ m}$    | 1                           |
| $R_\odot$ (solar radius)          | $6.96 \times 10^8 \text{ m}$    | $\times 4.3$                |
| $r_q$ (proplyd equilibrium)       | $1.45 \times 10^{10} \text{ m}$ | $\times 90$                 |
| $R_{s,\text{GR}}$ (Schwarzschild) | $2.95 \times 10^3 \text{ m}$    | $\times 5.5 \times 10^{-5}$ |

The BSFG horizon is  $\sim 55,000$  times larger than the GR Schwarzschild radius — but lies inside the stellar interior ( $0.233 R_\odot$ ), so it is only relevant for compact objects.

### §4 Surface Gravity and Hawking Temperature

**Step 3.** BSFG surface gravity (from metric derivative at  $r_h$ ):

$$\kappa_{\text{BSFG}} = \frac{c^2}{2} \left| \frac{\partial A_{00}}{\partial r} \right|_{r_h} = \frac{c^2}{2} \cdot \frac{3\eta |C_{\text{num}}| |\cos(\pi t_n)|}{r_h^4}$$

Using  $r_h^3 = \eta |C_{\text{num}}| |\cos|$ :

$$\kappa_{\text{BSFG}} = \frac{3c^2}{2r_h} \approx \frac{3 \times (3 \times 10^8)^2}{2 \times 1.62 \times 10^8} \approx 8.33 \times 10^8 \text{ m s}^{-2}$$

**Step 4.** BSFG Hawking temperature:

$$T_H^{\text{BSFG}} = \frac{\hbar \kappa_{\text{BSFG}}}{2\pi k_B c} = \frac{1.055 \times 10^{-34} \times 8.33 \times 10^8}{2\pi \times 1.381 \times 10^{-23} \times 3 \times 10^8} \approx 3.37 \times 10^{-12} \text{ K}$$

For comparison, the GR Hawking temperature for a solar-mass black hole:

$$T_H^{\text{GR}}(M_\odot) = \frac{\hbar c^3}{8\pi G M_\odot k_B} \approx 6.17 \times 10^{-8} \text{ K}$$

The BSFG Hawking temperature is  $\sim 18,000$  times colder than the GR result — consistent with a much larger horizon radius.

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## §5 Physical Interpretation

1. **The BSFG horizon is phase-dependent.** It only exists during the Aether anti-phase ( $\cos(\pi t_n) < 0$ ), appearing and disappearing on the Aether oscillation timescale. This “blinking horizon” has no GR analog.
  2. **The horizon lies inside stellar matter.** For a solar-type star,  $r_h \approx 0.23 R_\odot$ . The BSFG horizon is only physically accessible for compact objects where the stellar radius  $r_* < r_h$ , requiring a density  $\rho_* > 3M_\odot/(4\pi r_h^3) \approx 5.6 \times 10^5 \text{ kg m}^{-3}$  — white dwarf density range.
  3. **Distinct from  $r_q$ .** The proplyd equilibrium radius  $r_q \approx 0.097 \text{ AU}$  (PAPER\_550) is where  $U_m = 0$  — a force equilibrium in the DPM field. The BSFG horizon is where the metric degeneracy condition  $A_{00} = 0$  is met — a purely geometric criterion. The two coincide only by fine-tuning of Aether parameters.
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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                                                                                                  | SM / Experiment                 | Source               | Alignment                          |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------|------------------------------------|
| GR Schwarzschild metric recovery | BSFG line element<br>$\rightarrow g_{tt} = -(1-2GM/rc^2) \equiv \text{GR}$<br>in $\varepsilon_{\text{BSFG}} \rightarrow 0$ limit | Schwarzschild metric (GR exact) | PDG<br>2024 /<br>MTW | $\square$ BSFG<br>reduces to<br>GR |

| Observable                               | UQFF Prediction                                                                                                                   | SM / Experiment                                                  | Source                | Alignment                                         |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------|---------------------------------------------------|
| Shapiro time delay                       | BSFG geodesic $\rightarrow$<br>$\Delta t_{\text{BSFG}} \approx \Delta t_{\text{GR}} \times (1 + \varepsilon_{\text{correction}})$ | Cassini: $\Delta t / \Delta t_{\text{GR}} = 1 \pm 2.3\text{e-}5$ | Cassini/GR 2003       | □ Within Shapiro bound                            |
| Gravitational wave speed $v_{\text{GW}}$ | BSFG: $v_{\text{GW}} = c \times (1 + \kappa \eta^2) \approx c + 10^{-226} \text{ m/s}$                                            | GW150914 / GW170817:                                             | $v_{\text{GW}}/c - 1$ | $< 10^{-15}$                                      |
| Perihelion precession (Mercury)          | BSFG adds buoyancy correction $\delta\varphi = \kappa \times \varphi_{\text{GR}} \sim 10^{-6} \text{ arcsec/century}$             | GR prediction: $43.03''/\text{century}$ ; observed: $43.1''$     | GR + obs.             | UQFF correction undetectable at current precision |

**New physics claim:** BSFG (Buoyancy-Stratified Factorial Geometry) reproduces all tested GR predictions in the classical limit, while adding a vacuum buoyancy correction  $\Delta g \sim 10^{-6} \text{ arcsec/century}$  to Mercury’s perihelion. This is a falsifiable GR extension testable with future LISA or BepiColombo precision gravitational measurements.

*Cite PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

## §6 References

- CP4 #149 — BSFGRiemannCurvatureAetherMetricCalculator — PAPER\_554 ( $A_{00}(r)$  metric)
- CP4 #147 — Um26DPolyQuantizationDPMConfinementCalculator — PAPER\_550 ( $r_g = 0.097 \text{ AU}$ )
- CP4 #150 — BSFGGeodesicMetricCompatibilityCalculator — PAPER\_555 (geodesic equation)
- bh\_thermodynamics\_module.py — Hawking temperature framework (GR comparison)